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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

5254298087

BIOLOGY **5090/23**

Paper 2 Theory **May/June 2010**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

Section B
Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.
Write your answers in the spaces provided on the Question Paper.
Write an **E** (for Either) or an **O** (for Or) next to the number 8 in the Examiner's grid below to indicate which question you have answered.

You are advised to spend no longer than one hour on Section A and no longer than 45 minutes on Section B.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
Section B	
6	
7	
8	
Total	

This document consists of **14** printed pages and **2** blank pages.

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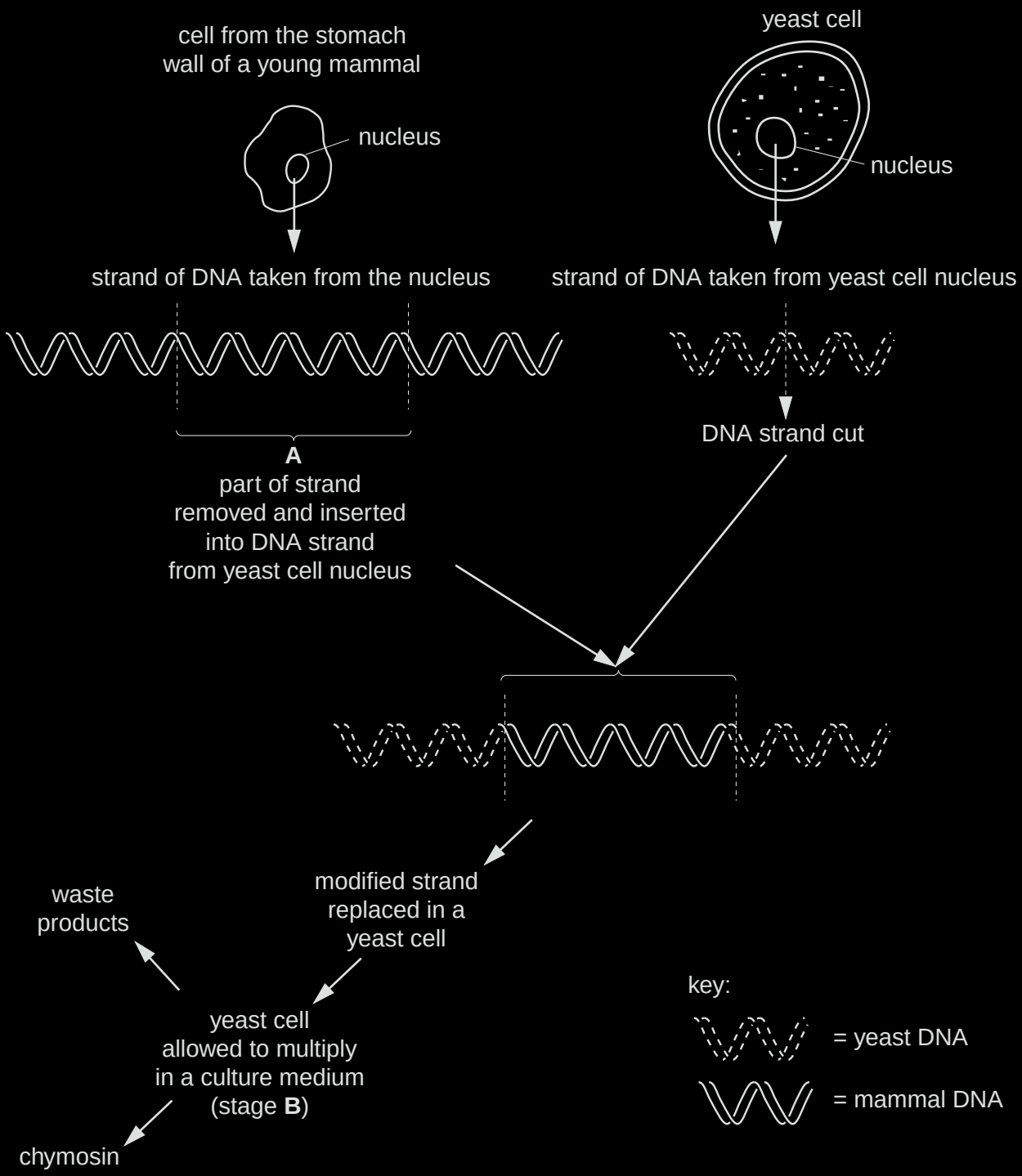


Fig. 1.1

(b) (i) Name the thread-like structure of which the DNA strand is a part.

..... [1]

(ii) Name the part of the strand labelled **A** in Fig. 1.1.

..... [1]

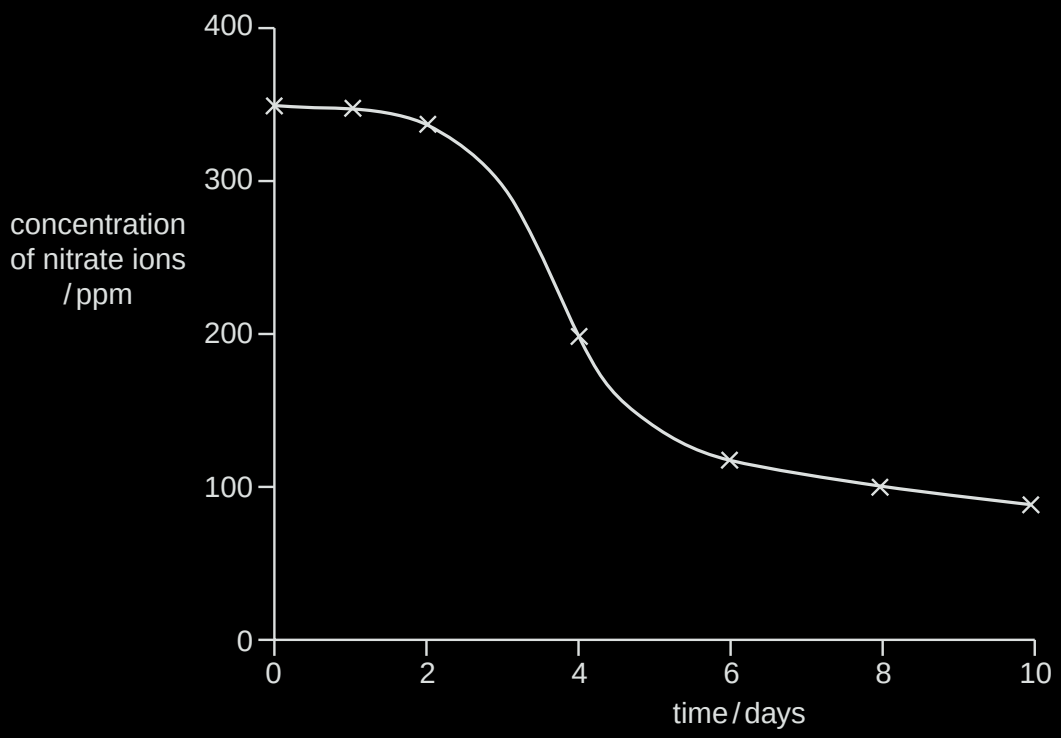


Fig. 2.1

(a) Use the graph to estimate the concentration of nitrate ions on day 5.

..... [2]

The plants in the container grew at their fastest rate between days 2 and 5.

(b) **(i)** Explain why they grew fastest during this time.

.....

.....

..... [3]

(ii) Suggest **one** way in which the rate of growth could have been further increased during this time.

..... [1]

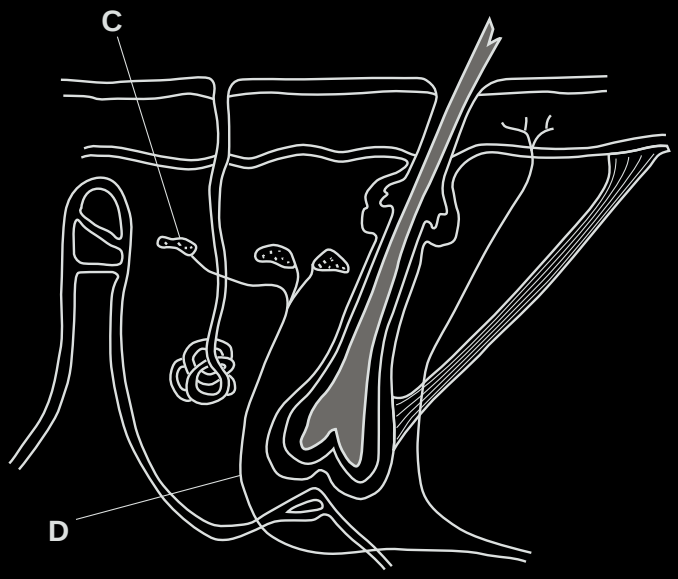


Fig. 4.1

(a) The body is able to maintain its internal environment within narrow limits. State the term for this process.

..... [1]

(b) Structures **C** and **D**, in Fig. 4.1, are involved in the process of temperature regulation.

Identify structures **C** and **D** and state the part they play in the process.

C .....

part played .....

.....

D .....

part played .....

..... [4]

(c) The consumption of alcohol causes the muscles in artery walls to relax.

Taking this into consideration, suggest why people who work in environmental temperatures below 5 °C might be advised not to drink alcohol before or during work.

.....

.....

.....

..... [3]

[Total: 8]

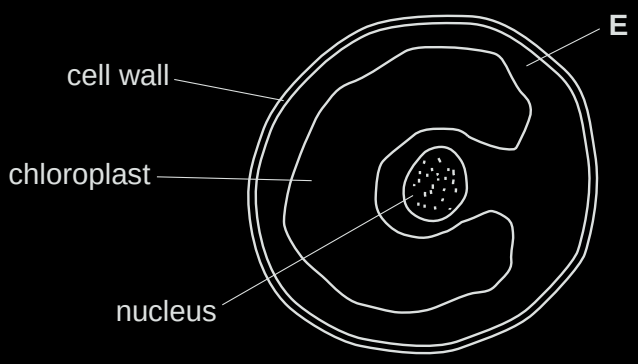


Fig. 5.1

(a) Suggest the identity of the region labelled **E** in Fig. 5.1.

..... [1]

(b) **(i)** List the structures visible in Fig. 5.1 which suggest that this organism might be a plant.

.....

..... [2]

(ii) List the ways in which this cell differs from a palisade cell in a plant leaf.

.....

.....

.....

..... [4]

